

Компьютерные сети (Д31). Якубова Ксения

Задание 1

568	6.994220	172.24.28.112	128.119.245.12	HTTP	672	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
578	7.188662	128.119.245.12	172.24.28.112	HTTP	305	HTTP/1.1 304 Not Modified
805	13.242128	172.24.28.112	128.119.245.12	HTTP	672	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
812	13.440038	128.119.245.12	172.24.28.112	HTTP	305	HTTP/1.1 304 Not Modified

1. Браузер использует версию HTTP версии 1.1 (как и на сервере)
2. Accepted Language – ru, en. Другая информация:

```
Host: gaia.cs.umass.edu\r\n
Connection: keep-alive\r\n
Cache-Control: max-age=0\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
Accept: text/html,application/xhtml+xml,application/xml;q=0.
Accept-Encoding: gzip, deflate\r\n
Accept-Language: ru,en;q=0.9\r\n
If-None-Match: "80-62e652d05f016"\r\n
If-Modified-Since: Tue, 18 Feb 2025 06:59:02 GMT\r\n
```

3. 172.24.28.112 – IP компьютера, 128.119.245.12 – адрес сервера.
4. HTTP/1.1 304 Not Modified
5. 18 Feb 2025 06:59:02 GMT
6. 305

Задание 2

107	9.953186	172.24.28.112	128.119.245.12	HTTP	561	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
116	10.144793	128.119.245.12	172.24.28.112	HTTP	796	HTTP/1.1 200 OK (text/html)
317	19.972279	172.24.28.112	128.119.245.12	HTTP	673	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
328	20.165688	128.119.245.12	172.24.28.112	HTTP	306	HTTP/1.1 304 Not Modified

1. Нет, не вижу
2. Да, вернул явно

```
\n
<html>\n
\n
Congratulations again! Now you've downloaded the file lab2-2.html. <br>\n
This file's last modification date will not change. <p>\n
Thus if you download this multiple times on your browser, a complete copy <br>\n
will only be sent once by the server due to the inclusion of the IN-MODIFIED-SINCE<br>\n
field in your browser's HTTP GET request to the server.\n
\n
</html>\n
```

3. Да, вижу. Это отметка времени позволяет клиенту указать серверу, что он хочет получить ресурс только в том случае, если он был изменен после указанной даты и

времени

If-Modified-Since: Tue, 18 Feb 2025 06:59:02 GMT\r\n

- Вернулось 304 (Not Modified). Сервер не вернул содержимое явно.

Задание 3

http						
No.	Time	Source	Destination	Protocol	Length	Info
895	10.787423	172.23.20.129	128.119.245.12	HTTP	561	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
906	10.932271	128.119.245.12	172.23.20.129	HTTP	874	HTTP/1.1 200 OK (text/html)

- 895
- 906
- Таких сегмента 3 (не включая последний запрос): их видно по [TCP PDU reassembled in 906]

902	10.931262	128.119.245.12	172.23.20.129	TCP	66	80 → 52419 [ACK] Seq=1 Ack=496 Win=30080 Len=0 TSval=1060048049 TSecr=4275626321
903	10.932261	128.119.245.12	172.23.20.129	TCP	1417	80 → 52419 [ACK] Seq=1 Ack=496 Win=30080 Len=1351 TSval=1060048049 TSecr=4275626321 [TCP PDU reassembled in 906]
904	10.932265	128.119.245.12	172.23.20.129	TCP	1417	80 → 52419 [ACK] Seq=1352 Ack=496 Win=30080 Len=1351 TSval=1060048049 TSecr=4275626321 [TCP PDU reassembled in 906]
905	10.932269	128.119.245.12	172.23.20.129	TCP	1417	80 → 52419 [ACK] Seq=2703 Ack=496 Win=30080 Len=1351 TSval=1060048049 TSecr=4275626321 [TCP PDU reassembled in 906]
906	10.932271	128.119.245.12	172.23.20.129	HTTP	874	HTTP/1.1 200 OK (text/html)

- Да, есть:

TCP segment data (808 bytes)	
[4 Reassembled TCP Segments (4861 bytes): #903(1351), #904(1351), #905(1351), #906(808)]	
[Frame: 903, payload: 0-1350 (1351 bytes)]	
[Frame: 904, payload: 1351-2701 (1351 bytes)]	
[Frame: 905, payload: 2702-4052 (1351 bytes)]	
[Frame: 906, payload: 4053-4860 (808 bytes)]	
[Segment count: 4]	
[Reassembled TCP length: 4861]	
[Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a46174653a204672692c2032312046656220323032352031353a30383a303920474d540	

Задание 4

No.	Time	Source	Destination	Protocol	Length	Info
377	5.021823	172.23.20.129	128.119.245.12	HTTP	561	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
383	5.161690	128.119.245.12	172.23.20.129	HTTP	1367	HTTP/1.1 200 OK (text/html)
385	5.170392	172.23.20.129	128.119.245.12	HTTP	507	GET /pearson.png HTTP/1.1
397	5.304596	128.119.245.12	172.23.20.129	HTTP	975	HTTP/1.1 200 OK (PNG)
405	5.394243	172.23.20.129	178.79.137.164	HTTP	474	GET /8E_cover_small.jpg HTTP/1.1
409	5.537433	178.79.137.164	172.23.20.129	HTTP	237	HTTP/1.1 301 Moved Permanently

- 3 GET запроса.
Первый GET - <http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html>
Второй GET - <http://gaia.cs.umass.edu/pearson.png>
Третий GET - http://gaia.cs.umass.edu/8E_cover_small.jpg
- По порядку моих запросов – последовательно, но совсем не обязательно, что так будет при каждом запросе.

Задание 5

No.	Time	Source	Destination	Protocol	Length	Info
606	6.183285	172.23.20.129	128.119.245.12	HTTP	577	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
611	6.328147	128.119.245.12	172.23.20.129	HTTP	783	HTTP/1.1 401 Unauthorized (text/html)
4753	61.086112	172.23.20.129	128.119.245.12	HTTP	662	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
4790	61.228113	128.119.245.12	172.23.20.129	HTTP	556	HTTP/1.1 200 OK (text/html)

- На начальное сообщение ответ 401 Unauthorized

2. Включается поле Authorization с логином и паролем

Cache-Control: max-age=0\r\n

✓ Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcm5=\r\n

Credentials: wireshark-students:network

Upgrade-Insecure-Requests: 1\r\n